

CASE STUDY · AUTOMATION ENGINEERING & TECHNICAL WRITING

AI Recruiting Automation Platform

An end-to-end LLM matching engine and a 14-service communications pipeline — designed, built, and documented from scratch, running in production.

AI Automation Engineer

Senior Technical Writer

LLM Systems · End-to-End

Wessam Mandour

THE PROBLEM

Recruiting top-of-funnel doesn't scale by hand

Hundreds of candidates per role. Multi-channel outreach under provider rate caps. Interview transcripts to score. And a final match a recruiter must actually trust.

Volume

Hundreds of applicants per role, most not a fit — manual screening is hours of work.

Rate limits

Email + WhatsApp outreach must respect provider caps and never double-send.

Transcripts

Screening interviews arrive as raw transcripts that need consistent scoring.

Trust

A bare match score is useless — a recruiter needs to know WHY a candidate ranked.

Two systems, one automated funnel

◆ Matching Engine

Python · FastAPI · ~3,900 LoC

Turns a free-text job description into a ranked shortlist with a written rationale per candidate, through a six-stage funnel.

◆ Candidate Flow

Node · Express · 14 services

The comms & data pipeline: batched email and WhatsApp outreach, transcript ingestion and scoring, and CV parsing — each isolated.

14

deployed services

6

stage funnel

100s

scored / run

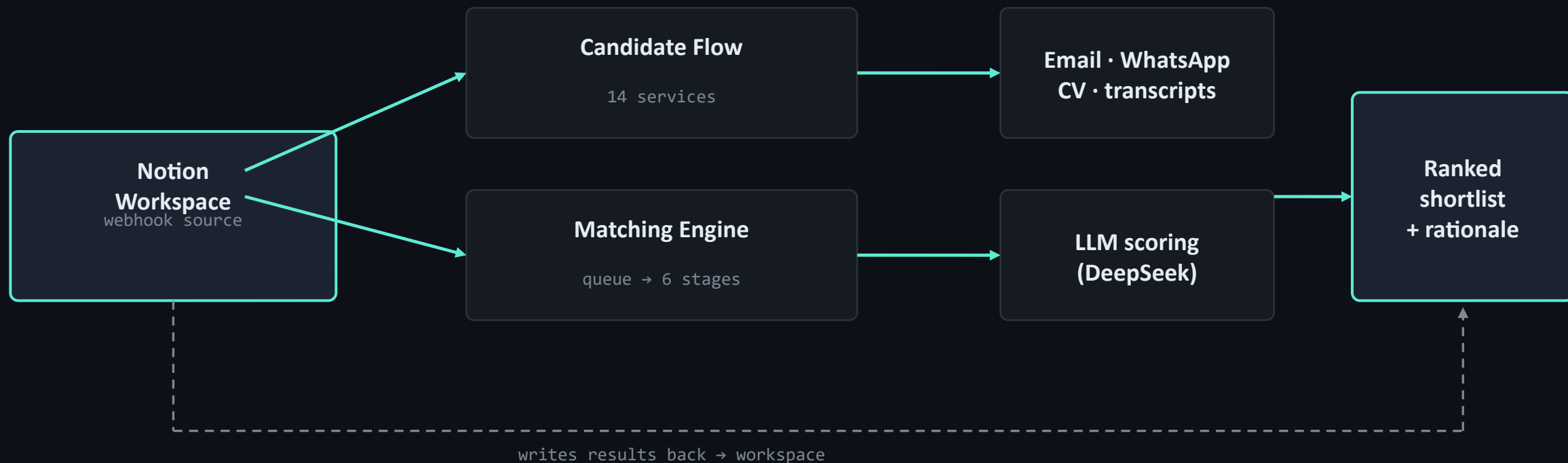
hrs→min

manual work cut

🚩 Placeholder figures — swap for your real, defensible numbers before presenting.

ARCHITECTURE

Event-driven, with the workspace as source of truth



No shared database to corrupt · failure in one service can't block another · every service stateless & idempotent

Six stages: job description → ranked shortlist

Each stage narrows the pool before the next, more expensive stage runs. The LLM only sees candidates worth paying to score.

1

Rubric extraction

LLM turns free-text job spec into a structured rubric: gates, requirements, keywords.

2

Keyword reduction

Cheap local filter drops no-signal candidates BEFORE any paid LLM call — top cost lever.

3

Hard filters

Non-negotiable gates: language, location, minimum screening score.

4

Priority ordering

Bucket by business priority; score the highest-value candidates first.

5

LLM scoring

0–100 + written rationale each, concurrency-capped, retried, failure-isolated.

6

Qualify & backfill

Link qualified first; backfill best below-threshold to hit target — logged.

The parts that make it production, not a script



Idempotent re-runs

Scored records skipped on re-run — no double-contact, no duplicate matches.



Rate limiting

Sequential queue, concurrency cap, write delays — under quota by design.



Graceful degradation

Backoff + JSON hardening; one bad candidate can't sink a 200-candidate run.



Schema-drift defense

Missing props ignored, odd types logged & skipped, eligibility re-checked.



Cost control

Cheap filters first, candidate caps, TTL cache — minimal LLM spend.



Observability

Logs keyed by id, email, stage, skip reason — debuggable from logs alone.

The documentation is part of the deliverable

Every service ships an operational handbook. For a system that runs unattended, the docs are the reliability layer.

Audience-layered

One-table overview for managers, branch-level depth for engineers — same doc.

Operational

Documents which log line proves which behavior: debug by grep, not by reading source.

Failure-first

Maps real symptoms to the exact gate that caused them.



troubleshooting.md

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## Troubleshooting
```

Candidate not considered for a role – check, in order:

1. Stage-2 keyword overlap
2. Stage-3 hard filters (language, location, score)
3. existing scored row (already processed)
4. duplicate-email de-duplication

Fewer qualified matches than target – expected:

Stage 6 backfills below-threshold by descending score and logs

`fallback_selected=true` per row.

What the automation delivers

100s

candidates scored per run, unattended

hrs → min

manual shortlisting per role, eliminated

14

integrations live with zero rate-limit incidents

1

explainable rationale on every single match

▀ *Replace these with your real figures — one concrete, honest number is what a hiring manager remembers.*

WHAT THIS DEMONSTRATES

Automation

14 event-driven services, idempotent, rate-limited, running unattended in production.

Technical writing

Audience-layered operational handbooks, failure-first troubleshooting, this case study.

Applied AI

A cost-aware, explainable, retried LLM scoring pipeline — not a single API call.

Wessam Mandour

wessam.mandour94@gmail.com · [linkedin.com/in/wessam-mandour](https://www.linkedin.com/in/wessam-mandour) · github.com/Wessam-Mandour